

Long HoDac

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EDUCATION

University Of California - San Diego

La Jolla, CA

Bachelor of Science in Computer Science

Expected Graduation: December 2026

- **Relevant Coursework:** Software Engineering, Computer Architecture, Machine Learning, Recommender Systems, Statistical Methods, NLP, Parallel Computing, Discrete and Continuous Optimization

EXPERIENCE

Workday - Payroll AI/ML Team

Pleasanton, CA

Incoming Software Engineer Intern

June 2026

UJIMA S&P Lab – UCSD CSE Department

La Jolla, CA

Software Engineer & Research Assistant

June 2025 – Present

- Constructed a data pipeline for 3,000+ TikTok videos comments using REST APIs, Pandas, and PostgreSQL
- Facilitated Agile events: stand-ups, sprint planning, spring review, and retro with 4 researchers over 6 weeks
- Applied NLP techniques(sentiment analysis, topic modeling) to 50,000+ comments revealing user behavior trends

NeuroML Lab – UCSD Cognitive Science Department

La Jolla, CA

Machine Learning Undergraduate Researcher

September 2025 – Present

- Extracted and L2-normalized fc7 embeddings from 50K+ ImageNet images via PyTorch + AlexNet batched GPU
- Implemented FAISS top-*k* nearest-neighbor search to flag “blind pairs” using cosine similarity threshold over 0.95
- Computed class-wise blind-pair rates and clusters to expose shape/texture failures, guiding experiments

House of Pancakes Academy - Non-Profit

San Diego, CA

Software Engineer Intern

October 2025 – December 2025

- Built a NFL player evaluation pipeline with 31K+ plays using Python and Bayesian models for the Big Data Bowl
- Developed a React + TypeScript dashboard highlighting six skill metrics to surface player strengths/weaknesses
- Engineered a Supabase pipeline processing 5M+ rows of records, enabling sub-second queries on 4K+ plays

De Anza College

Cupertino, CA

Intro To C++ Teaching Assistant

September 2023 – December 2023

- Guided 60+ students in C++ and development, providing debugging support and resolving logic/syntax errors
- Improved student exam scores by ~8% through targeted support and clarification of key programming concepts
- Created 5 programming problem sets using Notion to reinforce key concepts of object-oriented programming

PROJECTS

AI Anime Recommender with RAG | LangChain, Hugging Face, Chroma, Google Cloud, Kubernetes, Docker

- Developed Streamlit + LangChain recommender webapp delivering personalized suggestions via intuitive UI
- Built RAG backend with Hugging Face + ChromaDB, hitting 92% accuracy at 80 q/min via semantic search
- Deployed to Google Cloud Kubernetes via Docker and Helm with Grafana monitoring (<200ms latency)

Phishing Website Detection with MLOps | Scikit-learn, SciPy, FastAPI, MongoDB, MLflow, Dagshub, AWS S3

- Built an end-to-end ML pipeline for phishing/fraud detection, automating data ingestion, validation, and training
- Fine-tuned 5 classifier models with GridSearchCV on 11K URLs, achieving ~93% accuracy for phishing detection
- Implemented feature preprocessing with KNN imputation and transformers, improving F1 score by ~8–10%

Redis Chat platform | NextJS, TypeScript, Redis, TailwindCSS, Kinde Auth, Cloudinary

- Built a real-time chat app with Next.js, TypeScript, and Redis, delivering low-latency text/image messaging
- Designed Redis data models and Pusher channels to stream messages in order and eliminate client polling
- Implemented RBAC/OAuth2 with Kinde and Cloudinary uploads to securely handle user data and media at scale

TECHNICAL SKILLS

Languages: Python, Java, C/C++, TypeScript/JavaScript, SQL

Backend: Flask, FastAPI, Node.js, Express.js, REST APIs, Microservices, MongoDB, PostgreSQL

Frontend: React, Next.js, HTML/CSS, TailwindCSS, Bootstrap, Jest

AI/ML: PyTorch, scikit-learn, Pandas, NumPy, LangChain, Hugging Face, NLP (spaCy, NLTK), Prompt Engineering

Cloud/DevOps: AWS (S3, EC2, ECS), Google Cloud Platform, Docker, Kubernetes, Git, GitHub Actions (CI/CD)